A nuclear-spin quantum reservoir computer on real-world forecasting: pre-registered confirmation of non-inferiority, not superiority, to tuned classical models

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Abstract

Claims for quantum reservoir computing (QRC) rest mostly on synthetic benchmarks and on comparisons with weakly tuned classical models. We instead evaluate a simulated nine-spin ^{13}C crotonic-acid nuclear-magnetic-resonance (NMR) reservoir—exact Lindblad dynamics, the standard 653-bin free-induction-decay (FID) spectral readout ($D = 653$), linear ridge readout—on real-world forecasting, against a tuned echo-state network (ESN), a tuned LSTM and a tuned linear autoregressive (AR) model. Every dataset is split chronologically, every choice is made on validation data, and every design change is frozen and pre-registered in a version-controlled commit before it is confirmed on cities whose data had never been downloaded. On nine development datasets the best QRC variant (a learned input encoding) was never significantly better than the tuned ESN and was significantly worse on five. A memory analysis traced the long-horizon gap to input overwrite, and a single physical parameter—an encoding gain g that scales the input pulse angle, frozen at $g = 0.35$ on validation data—improved the pooled test R^2 over the standard encoding ($g = 1$) in all six cities where it was tested, a pooled-R^2 gain of $+0.015$ to $+0.079$. Five further pre-registered changes (bounded features, step time and compact readout, a learned encoding on top of the gain, a kernel readout, and ^{13}C or population readouts) were not adopted or failed confirmation; all are reported. The encoding of the closest experimental work scores below the frozen gain. In the second confirmation the frozen design is non-inferior to the tuned ESN, LSTM and linear AR in 3/3 cities (margin 0.02 in R^2, set after the first confirmation and disclosed). Over all nine untouched cities it is within the margin of the tuned ESN in 7/9, of the LSTM in 8/9 and of linear AR in 9/9, but never better than the ESN, which is significantly ahead in 6/9 (mean $\Delta R^2 = -0.011$; worst in Lagos, -0.049, where slow temperature anomalies demand long memory). The QRC fails under input distribution shift; a pre-registered test on two untouched, shifted series confirms that classical level handling (last-value re-centring plus raw input lags) removes most of the failure, but it gives the QRC no edge over the tuned classical models. On synthetic tasks a gradient-trained encoding improves the reservoir through its cross-step memory, yet a tuned ESN is one to three orders of magnitude better. We claim non-inferiority to tuned classical models on most untouched real-world data, and no quantum advantage.

I. INTRODUCTION

Reservoir computing (RC) trains only a linear readout on the state of a fixed nonlinear dynamical system, avoiding the cost and instability of training a recurrent core [1–4]. Quantum reservoir computing (QRC) replaces the classical reservoir with a quantum system whose large state space and intrinsic nonlinearity are conjectured to provide a rich temporal kernel at low training cost [9, 10, 15]. Nuclear-spin ensembles are an attractive QRC substrate: many coupled qubits, well-characterized Hamiltonians, decades of control technology, and a native, ensemble-averaged time-multiplexed observable—the free-induction decay (FID) [12, 19, 22].

Two features of the QRC literature make its practical value hard to judge. First, most evidence comes from synthetic benchmarks (NARMA, Mackey–Glass, memory capacity), where a well-tuned classical reservoir is a demanding reference. Second, the classical baseline is often weakly tuned. Our own synthetic results illustrate both points: on NARMA and Mackey–Glass a gradient-trained QRC encoding beats a tuned LSTM on two of three tasks, but a fully tuned ESN is one to three orders of magnitude better on all three (Sec. VIII). We therefore make real-world forecasting the primary test, and design the study so that its conclusion cannot be produced by selection on the test data.

The study has three stages. (i) A *development benchmark* on nine real-world datasets (daily weather, hourly air quality and transformer load, a chaotic laser), in which the standard QRC is compared with a tuned ESN, LSTM and linear AR model (Sec. IV). (ii) Six *design steps*, each pre-registered in a commit before it was run, developed on validation periods only and adopted only if it cleared a fixed threshold (Sec. V). (iii) Three *pre-registered confirmations* on nine cities, and a pre-registered *drift test* on two shifted non-weather series, whose data were downloaded only after the rules and success criteria were committed (Secs. VI and VII). A short section uses synthetic tasks, where ablations are cheap, to ask which parts of the reservoir carry the computation (Sec. VIII).

Our contributions are: (1) a leakage-controlled, pre-registered real-world forecasting protocol for a physical QRC, with all failures reported; (2) the finding that the long-horizon gap of an NMR reservoir is largely input overwrite, removable by one physical parameter (the encoding gain), confirmed out of sample in six untouched cities; (3) pre-registered evidence on nine untouched cities that the final design is mostly non-inferior to—but never

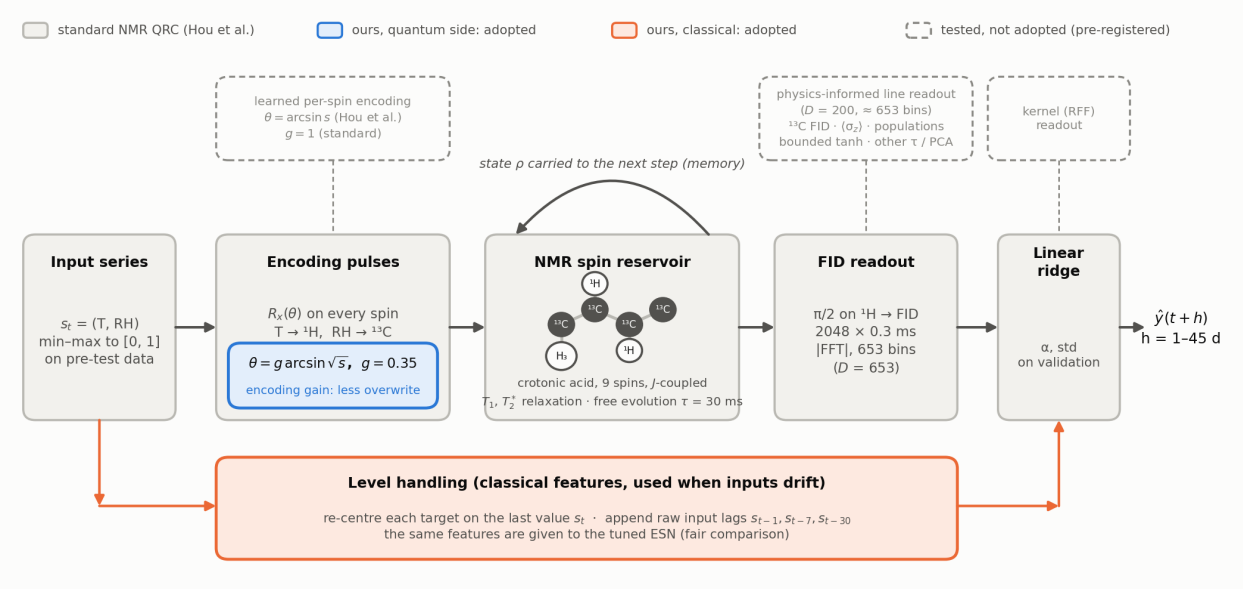


FIG. 1. The QRC forecasting pipeline. Each input step encodes the normalized series as $R_x(\theta)$ pulses (temperature on the protons, humidity on the carbons). The crotonic-acid reservoir then evolves freely for 30 ms, and its proton FID is read as 653 |FFT| bins ($D = 653$) into a linear ridge readout; the state carries over to the next step. Grey: the standard NMR QRC [22]. Blue: our adopted quantum-side change, the encoding gain $g = 0.35$ (Sec. V A; confirmed out of sample, Sec. VI). Orange: our adopted classical addition for drifting inputs, level handling (Secs. V E, VII), also given to the tuned ESN. Dashed: pre-registered changes tested and not adopted (Secs. V, V F).

better than—tuned classical models, with the remaining gap located in long memory; (4) a pre-registered drift test showing that the QRC's failure under input shift is fixed by classical level handling, not by the reservoir; and (5) a physics-informed readout derived from the Hamiltonian, which projects the FID onto its analytic J-multiplet lines. It matches the 653-bin readout with about 3× fewer features and is differentiable (Sec. II G).

II. THE PHYSICAL RESERVOIR AND ITS READOUT

Figure 1 summarizes the forecasting pipeline and marks what this work changed relative to the standard NMR QRC of Ref. [22], on the input side and on the output side, together with the pre-registered changes that were tested and not adopted.

A. Physical platform

We consider liquid-state NMR of an isotopically labeled molecule in a strong static field $B_0\hat{z}$. The sample is a macroscopic ensemble ($\sim 10^{18}$ molecules), so every observable is an ensemble average: the induced FID is a classical voltage proportional to the expectation value $\langle\hat{O}\rangle = \mathrm{Tr}\Big(\hat{O}\rho\Big)$, obtained without single-shot projection noise. This suits reservoir computing, where the readout is precisely a set of such expectation values. The reservoir for all real-world results is the nine-spin ^{13}C-labeled crotonic-acid molecule (four carbons, five protons, the three methyl protons magnetically equivalent) [22] (Table II). The mechanism study (Sec. VIII) additionally uses a seven-spin ^{13}C molecule (dccb7) and an abstract six-spin Ising reservoir.

B. Rotating-frame Hamiltonian

In the rotating frame of each spin's Larmor frequency and under the weak-coupling (secular) approximation valid when $|\nu_i - \nu_j| \gg |J_{ij}|$, the internal Hamiltonian reduces to a diagonal Ising form,

$$\frac{H}{h} = \sum_i \nu_i\, I_z^{(i)} \;+\; \sum_{i<j} J_{ij}\, I_z^{(i)} I_z^{(j)}, \qquad I_z = \tfrac{1}{2}\sigma_z, \tag{1}$$

with chemical-shift offsets ν_i (Hz) and scalar J-couplings J_{ij} (Hz); angular frequencies carry the usual factor 2π. All terms commute, so Eq. (1) is diagonal in the computational basis $\{|z\rangle\}$, $z \in \{\pm 1\}^n$, with eigenvalues

$$E(z)/h = \tfrac{1}{2}\sum_i \nu_i z_i + \tfrac{1}{4}\sum_{i<j} J_{ij} z_i z_j. \tag{2}$$

Entanglement is generated purely by the ZZ evolution acting on the transverse coherences that the input and readout pulses create; the diagonal structure makes the FID line structure analytic (Sec. II G).

C. Dissipation

Relaxation is modeled by a Lindblad master equation,

$$\dot{\rho} = -\frac{i}{\hbar}[H,\rho] + \sum_i \Big(\mathcal{D}[\hat{L}_i^{T_1}]\rho + \mathcal{D}[\hat{L}_i^{T_2}]\rho\Big), \quad \mathcal{D}[\hat{L}]\rho = \hat{L}\rho\hat{L}^\dagger - \tfrac{1}{2}\{\hat{L}^\dagger\hat{L},\rho\}, \tag{3}$$

with longitudinal (amplitude-damping) and pure-dephasing jump operators

$$\hat{L}_i^{T_1} = \sqrt{1/T_{1,i}}\,\sigma_-^{(i)}, \qquad \hat{L}_i^{T_2} = \sqrt{\gamma_{\phi,i}/2}\,\sigma_z^{(i)}, \quad \gamma_{\phi,i} = \frac{1}{T_{2,i}} - \frac{1}{2T_{1,i}}, \tag{4}$$

the physical (Bloch–Redfield-consistent) dephasing rate. We use measured T_1 and T_2^* for the crotonic system (Table II).

D. Initial state

At room temperature and NMR fields the spins are in the high-temperature limit; the equilibrium density matrix is, to leading order, $\rho_0 \propto \mathbb{1} + \epsilon \sum_i I_z^{(i)}$. We take the product form $\rho_0 = \bigotimes_i (\mathbb{1} + \epsilon\sigma_z^{(i)})/2$ with $\epsilon = 0.05$. Because Eqs. (3)–(4) are linear in ρ, the value of ϵ scales the overall signal amplitude but not the qualitative dynamics.

E. Input encoding and the encoding gain

A scalar input $s \in [0, 1]$ is written into the reservoir as a resonant radio-frequency rotation $R_x(\theta) = \exp\big(-i\,\theta \sum_i I_x^{(i)}\big)$ on the addressed spins. The standard encoding is $\theta(s) = \arcsin\sqrt{s}$, chosen so that a uniform input distribution maps to a uniform distribution of state overlaps [22]. We introduce an *encoding gain* $g \in (0, 1]$,

$$\theta(s) = g\,\arcsin\sqrt{s}, \tag{5}$$

so that $g = 1$ is the standard encoding. A smaller g rotates the stored state less on every step (less overwrite of the population memory) at the price of weaker input injection—the same trade-off that an ESN's input scale controls. For two-variable inputs (e.g. temperature and humidity) the first variable drives the protons and the second the carbons; a single-variable input drives every spin.

F. Free evolution, time multiplexing, and the standard FID readout

Each input step consists of (i) the encoding pulse $\rho \to U\rho U^\dagger$ and (ii) free evolution under Eq. (3) for a fixed time $\tau = 30\,\text{ms}$ (SOP standard). The readout is the proton FID, the transverse magnetization

$$M^+(t) = \sum_{k \in \text{readout}} \big\langle \sigma_x^{(k)} \big\rangle(t) + i \big\langle \sigma_y^{(k)} \big\rangle(t), \tag{6}$$

sampled densely in time; the dense sampling is itself the time multiplexing that enlarges the feature space [10]. In the simulation the FID is read from a copy of the state after each step. The feature vector of a step is the magnitude of the FFT of the FID at the 653 largest-magnitude bins (fixed once), the readout of Ref. [22]; we call it the *653-bin FID readout*, $D = 653$. **All real-world results in this paper use this readout.** A linear ridge readout $W = (X^\top X + \alpha I)^{-1} X^\top y$ maps the features to every forecast target; whether the features are standardized and the penalty α are chosen on validation data (Sec. III).

G. Physics-informed line readout

The 653-bin readout selects FFT bins by magnitude. We also derive the readout from the Hamiltonian. Because H in Eq. (1) is diagonal, the FID Eq. (6) is a sum over *single-quantum* transitions: flipping readout spin k against a fixed configuration $z \in \{\pm 1\}^{n-1}$ of the other spins oscillates, from Eq. (2), at

$$f_k(z) = \nu_k + \tfrac{1}{2} \sum_{j \neq k} J_{kj}\, z_j. \tag{7}$$

Each resonance is split into a 2^{n-1}-line J-multiplet, and every peak of the 653-bin readout is one of these lines or an FFT skirt of one. The encoding sets only the line amplitudes; the frequencies are fixed by H. Merging lines closer than the decoherence linewidth $\Delta f = 1/(\pi T_2)$ leaves D_{eff} resolvable centres (Table I), which are read by a fixed direct-DFT projection

$$X_k = \sum_{v=0}^{V-1} M^+(t_v)\, e^{\,i2\pi f_k t_v} = \left(M^+ \Phi\right)_k, \qquad \Phi_{v,k} = e^{\,i2\pi f_k t_v}, \tag{8}$$

over V FID samples, keeping real and imaginary parts: $D = 2D_{\text{eff}}$ real features. This *line readout* has three properties. It counts the reservoir's actual output dimension instead of choosing it. It is a single matrix product, so it can be back-propagated through, which the learned encoding of regime B requires (Sec. III). And it keeps the phase that |FFT| discards. On real data at $g = 0.35$, crotonic-9's 100 proton lines ($D = 200$) reach a mean development validation R^2 of 0.7021 against 0.7059 for the 653 bins, with about 3× fewer features (Table VI). It was not adopted because the rule required a gain, but it is the readout of every mechanism result (Sec. VIII).

TABLE I. Physics-informed line count (SOP §3.2): single-quantum lines of the readout spins, the decoherence linewidth Δf, and the resolvable centres D_{eff} after merging lines closer than Δf. The line readout has $D = 2D_{\text{eff}}$ real features. Doubling the coupling ($c = 2$) widens the multiplets and lifts near-degeneracies.

system (readout spins)	raw lines	Δf (Hz)	D_{eff}
crotonic-9 (protons)	1280	1.57	100
dccb7 (seven ^{13}C)	448	0.60	172
abstract 6-spin, c=1	192	1.59	137
abstract 6-spin, c=2	192	1.59	161

H. Numerical methods

We vectorize the density matrix, $\rho \to \vec{\rho} \in \mathbb{C}^{d^2}$ ($d = 2^n$), and write Eq. (3) as $\dot{\vec{\rho}} = \mathcal{L}\vec{\rho}$. Because H is diagonal and the dissipators preserve coherence order, $\exp(\tau\mathcal{L})$ is exact and block-diagonal in coherence patterns; for $n \geq 7$ we use this exact block propagator, which makes multi-decade daily series tractable for nine spins. For small systems ($d^2 \leq 8192$) we precompute the dense node propagator, which is also differentiable (Sec. VIII).

I. Two training regimes

We distinguish, and label on every result, two regimes. In **regime A** (standard QRC) the reservoir and the encoding are fixed and only the linear readout is trained by one closed-form ridge solve. In **regime B** (learned encoding) the same ridge readout is used, but the input encoding is additionally trained by gradient descent through the differentiable reservoir. The final real-world design is regime A: the encoding gain g is a fixed hyperparameter, frozen on validation data, not a trained weight.

On hardware, regime B cannot back-propagate through the spins. Each encoding pulse $R_x(\theta_{t,i})$ acts on one spin, so its generator has two eigenvalues and the two-term parameter-shift rule [17, 18] gives the exact gradient from forward experiments alone: $\partial L/\partial\theta_{t,i} = \frac{1}{2}[G \cdot F(\theta + \frac{\pi}{2}e_{t,i}) - G \cdot F(\theta - \frac{\pi}{2}e_{t,i})]$, where F are the measured features and $G = \partial L/\partial F$ comes from the classical ridge readout. We trained a per-spin encoding this way, using only

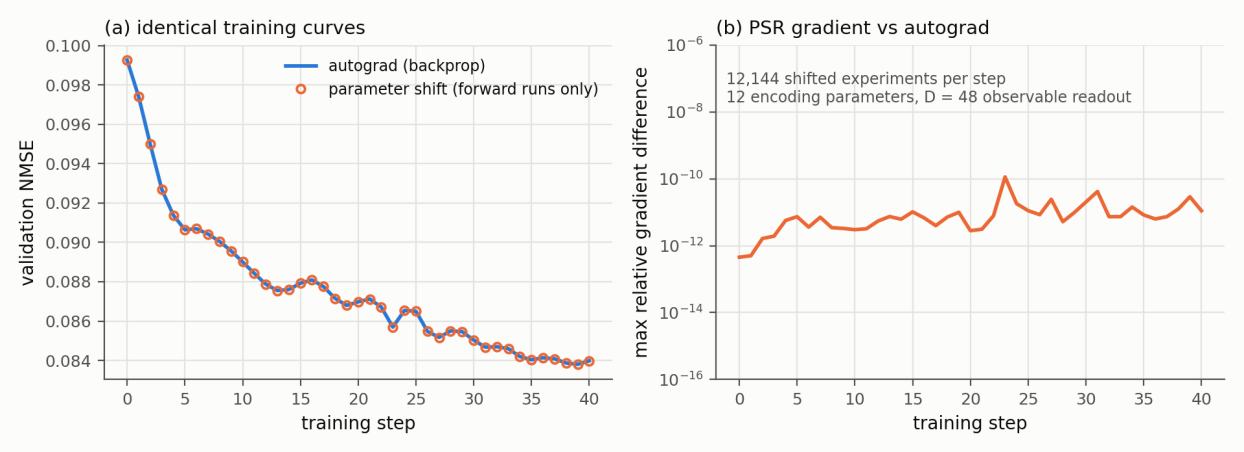


FIG. 2. Training the encoding with parameter-shift gradients (forward runs only, as on hardware) versus autograd, from the same start ($g = 0.35$, 12 parameters, four-spin test reservoir, observable readout $D = 48$, Delhi weather). (a) Validation NMSE per step: the curves coincide. (b) Largest relative difference between the two gradients at each step.

shifted forward runs, next to an identical autograd run (Fig. 2). The task is Delhi weather, next-day temperature and humidity, on a four-spin test reservoir with an observable readout ($D = 48$, not the 653-bin standard). The two training curves agree for all 40 steps; the gradients differ by at most 1.1×10^{-10} (relative). The cost is $2\,n\,T = 12{,}144$ experiments per gradient step ($n = 4$ spins, $T = 1518$ days). This validates the hardware training path. It is not a performance result: learned encoding was not adopted for the real-world design (Sec. V D).

III. BENCHMARK PROTOCOL

A. Datasets and chronological splits

Every dataset is split chronologically into train / validation / test blocks, fixed before any result. Normalization (min–max) uses pre-test data only. The nine *development* datasets are: daily temperature and relative humidity of Delhi from a Kaggle station record (the task of Ref. [22]; 300 training days, 500 test days); Open-Meteo ERA5 reanalysis daily temperature and humidity, 1970–2025, for Delhi, Mumbai, Bangkok, Beijing and Berlin (train 1971–1999, validation 2000–2004, test 2005 to the end of the record, 7625 test days);

TABLE II. Nine-spin ^{13}C crotonic-acid reservoir [22] used for all real-world results: chemical-shift offsets ν, relaxation times, and the symmetric J-coupling matrix (Hz). H3–H5 are the equivalent methyl protons. With $\tau = 30\,\mathrm{ms}$, transverse coherences (T_2^* 0.20–0.33 s) survive 7–11 steps and populations (T_1 2.2–27.5 s) 73–917 steps.

spin	ν (Hz)	T_1 (s)	T_2^* (ms)
C1	−7749.7	5.9	212
C2	5430.1	4.9	231
C3	2699.9	5.6	208
C4	7673.7	27.5	241
H1	985.9	3.2	203
H2	520.3	3.4	332
H3–H5	−1081.5	2.2	320

pair	J	pair	J	pair	J
C1–C2	40.8	C2–C3	69.5	C1–C3	1.6
C1–C4	8.5	C2–C4	1.4	C3–C4	71.0
C1–H1	4.0	C2–H1	155.6	C3–H1	−1.8
C4–H1	6.5	C1–H2	6.6	C2–H2	−0.7
C3–H2	162.9	C4–H2	3.3	H1–H2	15.8
C1–Me	128.0	C2–Me	−7.1	C3–Me	6.6
C4–Me	−0.9	H1–Me	6.9	H2–Me	−1.7

the hourly Beijing PM2.5 station record (temperature and dew point; train 2010–2012, validation 2013, test 2014); the ETTh1 transformer record (inputs oil temperature and load, target oil temperature, standard 12/4/4-month split); and the Santa Fe laser A chaotic series. Weather targets are each variable at horizons $h = 1, 5, 10, 15, 20, 30, 45$ days (14 targets); the other datasets use their own horizon sets. The nine *confirmation* datasets are further ERA5 cities with exactly the ERA5 protocol: Chennai, Cairo and Moscow (Confirmation 1), Sydney, São Paulo and Toronto (Confirmation 2), and Lagos, Mexico City and Stockholm (Confirmation 3), chosen for climatic diversity; ETTh2 serves as the untouched distribution-shift test of design step 2. The *drift test* (Sec. VII) uses two standard long-term forecasting

series: daily exchange rates 1990–2016 (target the first currency, second input the second; horizons 1, 5, 10, 20 days) and hourly electricity consumption 2012–2014 (target the last of 321 clients, second input the mean over all clients; horizons 1, 6, 12, 24, 48 h), each split chronologically 70/10/20%.

B. Models and validation-only selection

QRC: the crotonic-9 reservoir of Sec. II, 653-bin FID readout ($D = 653$), ridge readout with α and the standardization flag chosen on validation, refit on train+validation, test read once. *Tuned ESN*: leaky ESN, grid over size (≤ 1000) $\times$ spectral radius $\times$ leak $\times$ input scale (three reservoir seeds), selected on validation; a *size-matched* ESN restricts the size to ≤ 653, the QRC's feature count. *Tuned LSTM* [6]: hidden size $\times$ layers $\times$ learning rate, three initializations, early stopping on validation R^2, the selected configuration retrained on train+validation for the selected epoch count. *Tuned linear AR*: ridge regression on lagged inputs, lags and penalty chosen on validation. No model sees the test block before its configuration is fixed.

C. Metric and significance

The metric is the pooled test R^2, the mean over all forecast targets (variables $\times$ horizons); higher is better. Short horizons are $h \leq 10$ days and long horizons $h \geq 30$ days. Uncertainty is a 95% moving-block bootstrap over test time (2000 resamples; 30-day blocks in the confirmations), paired on the same test days, applied to the *difference* in pooled R^2 between two models. A difference is called significant when its CI excludes 0. Replicate s.d. (encoder, reservoir or initialization seeds) is reported where a model has replicates; the fixed-encoding QRC is deterministic.

D. Development, freezing, and pre-registration

The test blocks of the nine development datasets were scored once in the development benchmark (Sec. IV); from then on they are treated as *seen*. All subsequent design work (Sec. V) used development *validation* periods only. Each design step was written as a

pre-registration—the search space, a freeze rule with a numerical adoption threshold, and the success criteria of its confirmation—and committed to the project repository before it was run. Each confirmation's data were downloaded only after its pre-registration was committed. Table III lists the commits and outcomes; all outcomes, including failures, are reported.

IV. DEVELOPMENT BENCHMARK

Table IV gives the development benchmark at the standard encoding ($g = 1$), with the fixed encoding ("arcsin", regime A) and with a per-spin encoding correction trained on a training-period slice ("learned", regime B, three encoder seeds). The learned QRC improved on arcsin on 8 of 9 datasets, but against the tuned ESN it was never significantly better, comparable on 4 and significantly worse on 5; against the tuned LSTM it was better on 2 (the small Kaggle Delhi set and the noise-dominated Bangkok set), comparable on 1 and worse on 6. The highest mean R^2 was the QRC's on one dataset only (Kaggle Delhi, 300 training days, where it is comparable to the ESN). On ETTh1 the QRC fails outright (negative R^2).

Why the QRC loses where it loses. A post-hoc factor analysis over 101 forecast targets (ETTh1 excluded) identified forecast horizon as the main factor against the ESN: the QRC−ESN difference falls with horizon (Spearman -0.47), from -0.0001 at short horizons to -0.041 at long horizons. The memory function (test R^2 of reconstructing the input L steps back, ERA5 Delhi) makes this direct: the QRC reaches 0.92 / 0.77 / 0.53 at lags 10 / 30 / 90 days, the tuned ESN 0.97 / 0.94 / 0.88. The QRC's memory outlasts its coherence time (7–11 steps), so population memory already reaches the readout; the likely limiter is that every input pulse, up to $\pi/2$ per spin, partly overwrites the stored populations. In ESN language, the standard QRC runs at a large, fixed input scale. Two further factors: the QRC beats the LSTM only where the LSTM is data- or signal-starved, and on ETTh1 the test period shifts regime (inputs 1.31 s.d. from the training mean) so that the unbounded FFT features leave their training range (up to 16–29 s.d.), whereas the ESN's bounded states degrade gracefully. A tuned linear AR model is within 0.01–0.04 R^2 of the best nonlinear model on most weather targets, and is the best model on ETTh1. These observations motivated the design concepts of Sec. V; because they were made on seen test data, nothing in them is claimed until confirmed on untouched data.

TABLE III. Pre-registered steps, in order. "Commit" is the repository commit that fixed the rule and criteria before the step was run (and, for confirmations, before the data were downloaded). Dev = mean pooled validation R^2 over the eight counted development datasets.

step	pre-registration commit	freeze / success rule	outcome
1: encoding gain g (C1)	`a1dfb00` (hypotheses H1, H2)	H1: memory rises as g falls; H2: long-horizon gain without short-horizon loss	H1 8/8, H2 6/8; $g = 0.35$ frozen
Confirmation 1	`29152bf`	S1: $\Delta(g{=}0.35 - g{=}1) > 0$, CI excl. 0, $\geq 2/3$ cities; S2: long $\geq +0.0205$	S1 pass 3/3; S2 pass
2: bounded features (C6)	`53d8e6a` (transform frozen `34156a7`)	R1 on untouched ETTh2: $R^2 \geq 0$ and not worse than raw	R1 fail; not supported
3: step time τ + compact readout (C5, C7)	`ed96b17`	adopt only if Dev $\geq$ current +0.005	not adopted (best = current)
4: learned encoding on top of g	`5b92610`	adopt only if Dev $\geq$ fixed $g = 0.35$ +0.005	not adopted ($\Delta = -0.0021$)
Confirmation 2 (final design)	`7965cde`	F1: replicate S1; F2: non-inferiority, margin $\delta = 0.02$, per baseline	F1 pass 3/3; F2 pass 3/3 (ESN, LSTM, AR)
5: readout-side variants (R1–R5)	`a7cb3e6`	adopt best if Dev $\geq$ base +0.005, worse on $\leq 2/8$	kernel (RFF) adopted
Confirmation 3 (kernel readout)	`a7cb3e6`	G1: Δ(kernel − base) > 0, CI excl. 0, $\geq 2/3$; G2: non-inferior to ESN-kernel, $\geq 2/3$	G1 fail 0/3; G2 fail 1/3; withdrawn
Drift test (level handling)	`a4111de`	D1: Δ(level − base) > 0, CI excl. 0, on both series	D1 pass 2/2
6: paper encoding; ^{13}C / population readout	`d022533`	as step 5, plus gain must survive dropping the best dataset	not adopted; Conf. 4 not run

TABLE IV. Development benchmark, pooled test R^2 (mean ± s.d. over three replicates; QRC arcsin is deterministic). QRC: crotonic-9, $g = 1$, $\tau = 30\,\mathrm{ms}$, 653-bin FID readout ($D = 653$). Significance of learned QRC vs (arcsin, tuned ESN, matched ESN, tuned LSTM): + significantly better, − significantly worse, ≈ 95% CI contains 0.

dataset	test	QRC arcsin	QRC learned	ESN	ESN ≤ 653	LSTM	lin. AR	sig.
Delhi (Kaggle)	500 d	0.694	$\mathbf{0.731 \pm 0.002}$	0.716 ± 0.001	0.717 ± 0.001	0.416 ± 0.081	0.689	+ ≈ ≈ +
Delhi (ERA5, daily)	7625 d	0.747	0.755 ± 0.000	0.765 ± 0.003	0.765 ± 0.003	$\mathbf{0.769 \pm 0.004}$	0.737	+ − − −
Mumbai (ERA5)	7625 d	0.600	0.621 ± 0.001	$\mathbf{0.706 \pm 0.003}$	0.706 ± 0.003	0.694 ± 0.015	0.666	+ − − −
Bangkok (ERA5)	7625 d	0.297	0.322 ± 0.000	$\mathbf{0.333 \pm 0.002}$	0.332 ± 0.007	0.272 ± 0.020	0.299	+ ≈ ≈ +
Beijing (ERA5)	7625 d	0.638	0.647 ± 0.002	0.649 ± 0.001	0.649 ± 0.001	$\mathbf{0.651 \pm 0.004}$	0.635	+ ≈ ≈ ≈
Berlin (ERA5)	7625 d	0.563	0.600 ± 0.000	0.604 ± 0.002	0.604 ± 0.002	$\mathbf{0.616 \pm 0.008}$	0.605	+ ≈ ≈ −
Beijing PM2.5	8712 h	0.901	0.914 ± 0.002	0.931 ± 0.000	$\mathbf{0.931 \pm 0.000}$	0.926 ± 0.003	0.930	+ − − −
ETTh1	2832 h	−0.238	-2.939 ± 3.992	0.662 ± 0.009	0.574 ± 0.015	0.250 ± 0.111	**0.732**	− − − −
Santa Fe laser A	2383	0.910	0.918 ± 0.008	0.958 ± 0.005	0.958 ± 0.005	$\mathbf{0.971 \pm 0.003}$	0.717	+ − − −

V. DESIGN STEPS (VALIDATION DATA ONLY)

Each step below was pre-registered (Table III) and developed on the validation periods of the eight counted development datasets (Kaggle Delhi, with only 300 training days and a 60-day inner split, is reported separately as a small-data check). No test block is used in any step except the untouched ETTh2 confirmation of step 2 and a labelled post-hoc check of step 5 on the already-seen ETT test sets.

A. Step 1: encoding gain

We swept $g \in \{0.1, 0.2, 0.35, 0.5, 0.75, 1\}$ in Eq. (5) (regime A, no training). Lowering g raised the memory function at a mid lag on all eight datasets (H1 supported, 8/8) and improved long-horizon validation R^2 by at least half the benchmark gap (≥ 0.0205) without losing short horizons on six of eight (H2 met, 6/8) (Table V). Because "best g" is chosen and scored on the same validation period, the per-dataset optimum is optimistic; the robust finding is the trend. The frozen rule is a single fixed gain, $g = 0.35$: it has the best mean rank over the eight datasets (2.12 of 6 gains), the best mean long-horizon validation R^2 (0.618), ties the best mean pooled validation R^2 (0.700), and beats $g = 1$ on all eight. At $g = 0.1$ the readout becomes ill-conditioned on small data (Kaggle Delhi validation R^2 down to −6.3), so the best gain is data-regime dependent. Figure 3 shows the mechanism: lowering g from

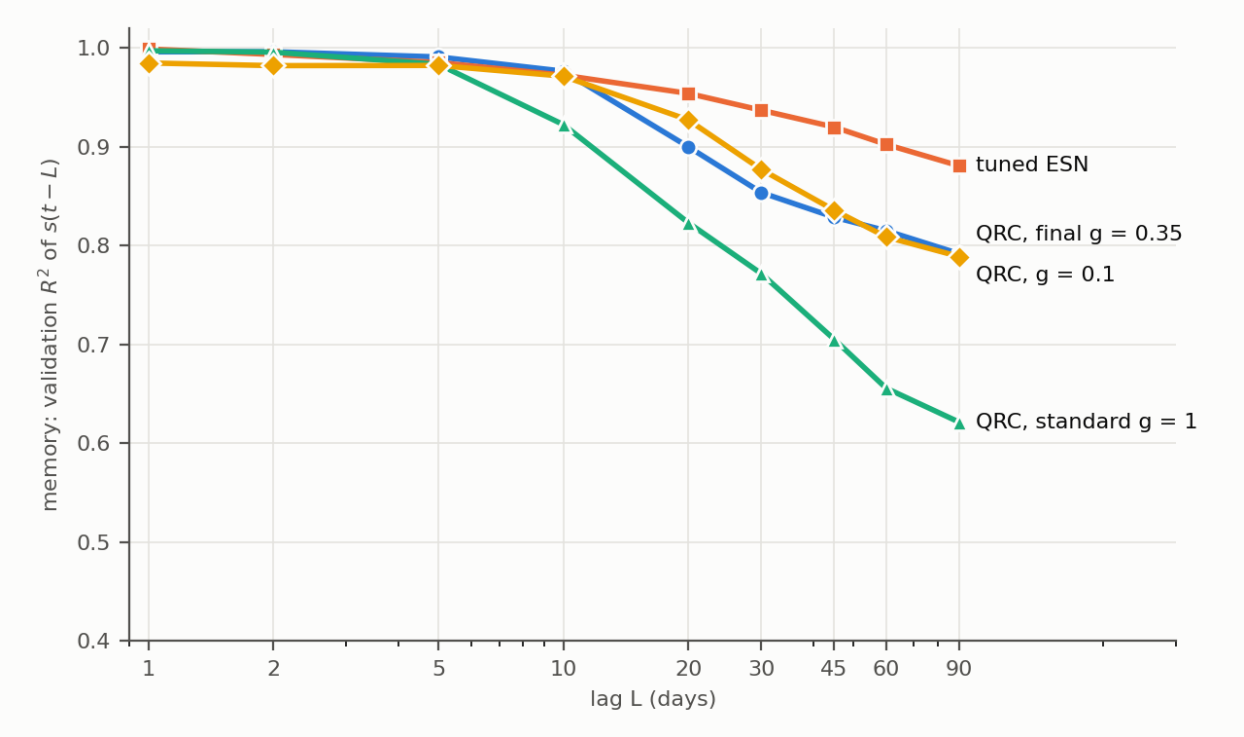


FIG. 3. Memory function on ERA5 Delhi: validation R^2 of reconstructing the input L days back from the current features, for the QRC (crotonic-9, $\tau = 30$ ms, 653-bin FID readout, $D = 653$) at three encoding gains and for the tuned ESN (same function, same split). Lowering the gain reduces input overwrite; the remaining gap to the ESN is long memory.

1 to 0.35 lifts the memory of the input 90 days back from 0.62 to 0.79, but no gain reaches the tuned ESN (0.88); below $g = 0.35$ nothing more is gained.

B. Step 2: bounded readout features (not supported)

To address the ETTh1 drift failure, three bounded transforms of the 653 features (clipped z-score, $\tanh(z/3)$, training-set rank), fitted on training data, were compared on development validation at $g = 0.35$; $\tanh(z/3)$ was frozen (mean 0.7067 vs raw 0.7061) before the untouched ETTh2 series was downloaded. On ETTh2 the pre-registered criterion failed: pooled test R^2 was $+0.2914$ with the transform vs $+0.3632$ raw (difference -0.0717, CI $[-0.1880, +0.0320]$), against $+0.7911$ (tuned ESN), $+0.7467$ (tuned LSTM) and $+0.8113$ (linear AR). The drift failure replicates out of sample for the standard QRC ($g = 1$: $R^2 = -0.1506$). The cause is extrapolation, not feature size: under drift the inputs leave

TABLE V. Step 1 (validation only): pooled validation R^2 at the frozen $g = 0.35$ vs the standard $g = 1$, and the memory function (validation R^2 of reconstructing the input at lag 20, or 24 for hourly data) at $g = 0.1$, 0.35, 1. QRC: crotonic-9, $\tau = 30$ ms, 653-bin FID readout ($D = 653$).

	pooled val. R^2		memory function		
dataset	$g = 1$	$g = 0.35$	$g = 0.1$	$g = 0.35$	$g = 1$
ERA5 Delhi	0.746	0.758	0.93	0.90	0.82
ERA5 Mumbai	0.641	0.714	0.92	0.89	0.77
ERA5 Bangkok	0.222	0.294	0.86	0.81	0.54
ERA5 Beijing	0.628	0.638	0.81	0.79	0.69
ERA5 Berlin	0.538	0.579	0.83	0.80	0.67
Beijing PM2.5 (hourly)	0.920	0.923	0.98	0.97	0.95
ETTh1 (hourly)	0.788	0.814	0.83	0.88	0.79
Santa Fe	0.762	0.880	0.99	0.97	0.80

the training range, where the QRC's strongly nonlinear feature map gives the linear readout no reliable relation. The frozen gain lifts the QRC from -0.15 to $+0.36$ on ETTh2 (not pre-registered for ETTh2), still about 0.4 below the classical models. This is a documented limitation.

C. Step 3: step time and compact readout (not adopted)

At $g = 0.35$ we scored $\tau \in \{15, 30, 60, 100\}$ ms $\times$ six readouts (653, 200 and 100 top bins; PCA to 128 and 64; the physics-informed line projection of Sec. II G) on development validation (Table VI). The current design ($\tau = 30$ ms, 653 bins, mean 0.7059) was the best of all 24 combinations, so nothing was adopted and no confirmation was run. The optimum in τ is flat once the gain is set; the line projection is near-equal (0.7021) with $\sim 3\times$ fewer features, and on the small Kaggle Delhi set compact readouts help slightly (0.590 vs 0.582 at 30 ms).

TABLE VI. Step 3 (validation only): mean pooled validation R^2 over the eight development datasets at $g = 0.35$. Readout labels: bins$K$ = top-K FFT bins of the FID ($D = K$; bins653 is the SOP standard); pca$K$ = PCA to K components fitted on training data; lines = physics-informed line projection (reduced FID). Bold: best.

τ (ms)	bins653	bins200	bins100	pca128	pca64	lines
15	0.7008	0.6962	0.6841	0.6948	0.6846	0.7002
30	**0.7059**	0.7028	0.6966	0.7028	0.6954	0.7021
60	0.7027	0.7014	0.6969	0.7000	0.6945	0.7022
100	0.6978	0.6983	0.6973	0.6953	0.6894	0.7000

D. Step 4: learned encoding on top of the gain (not adopted)

We trained a per-spin correction $\theta = 0.35 \arcsin\sqrt{s} + 0.5 \tanh(\mathrm{MLP}(s))$, zero-initialized so that training starts at the current design, with the recipe of the development benchmark (training-period slice only, 30 Adam steps, seeds 0–2; regime B, 653-bin readout). Its seed-mean validation R^2 was 0.7038 vs 0.7059 for the fixed gain ($\Delta = -0.0021$, threshold $+0.005$; better on 4 of 8), so it was not adopted. The fixed $g = 0.35$ also beats the published learned encoding at $g = 1$ on 6 of 8 datasets (mean 0.7059 vs 0.6927). On real data, therefore, the learned-encoding gain of the development benchmark is largely explained by reduced input overwrite, and the two improvements are not complementary: once g is reduced, the learned correction mostly fits the training slice without transferring to validation.

E. Step 5: readout-side variants (kernel adopted, then withdrawn)

Five readout-side variants were added on top of the 653 features and given identically to the tuned ESN (validation-selected hyperparameters): last-value re-centring ("delta"), raw input lags ("lags", and both together, "level handling"), a ridge fit to the residual of the tuned linear AR, delay-embedded feature principal components (classical memory, labelled as such), and a random-Fourier-feature kernel readout (2000 features). By the pre-registered rule the kernel readout was adopted (mean development validation R^2 0.7115 vs 0.7061; worse on 2/8), but the whole gain came from Santa Fe (+0.059); without it the

change was −0.002. It then failed its confirmation (Sec. VI C) and is withdrawn. A post-hoc check on the seen ETT test sets, labelled as such, showed the kernel readout collapsing under drift and level handling helping (ETTh2 +0.36 → +0.74, ETTh1 −0.21 → +0.23); that observation is what the drift test of Sec. VII confirms on untouched data.

F. Step 6: published encoding and population readouts (not adopted)

Two physical changes remained after step 5. (A) The encoding of the closest experimental work, $\theta = \arcsin s$ [22], as a positioning control. (B) Reading the long-lived ^{13}C spins (T_1 up to 27.5 s) as well as the protons: a ^{13}C FID (653 bins), the nine single-spin $\langle\sigma_z\rangle$, or all 512 computational-basis populations (idealised; population tomography on hardware), each concatenated with the proton readout ($D = 662$–1306). The adoption rule was that of step 5, plus a guard learned there: the gain must survive dropping the single most-improved dataset. Nothing was adopted (Table VII). The published encoding scored 0.6903 against 0.7060 for $g = 0.35$ (worse on 5/8; better than $\arcsin\sqrt{s}$ at $g = 1$, 0.6798, since $\arcsin s \leq \arcsin\sqrt{s}$ also rotates less). The best readout, proton $\oplus$ ^{13}C FID, gained +0.0023, almost all of it on Santa Fe, and no channel changed the long-lag memory on any weather city (Delhi, lag 90: 0.79 for every channel). The memory gap is therefore not a readout limit: the proton J-multiplets already resolve the carbon configurations, and since H is diagonal the populations change only through the input pulses and T_1. Memory is set by how each pulse overwrites the stored state, which no readout can undo.

Final design. Steps 1–6 leave the final design: crotonic-9, fixed gain $g = 0.35$ (no learned correction), $\tau = 30$ ms, 653-bin FID readout ($D = 653$), ridge readout. It is regime A: no gradient enters the quantum dynamics. Where inputs can drift, level handling is added to the readout as classical features (Sec. VII).

VI. PRE-REGISTERED CONFIRMATIONS ON UNTOUCHED CITIES

A. Confirmation 1: the gain rule

Pre-registered in commit `29152bf` before the data were downloaded, Confirmation 1 tested the frozen gain on Chennai, Cairo and Moscow (test 2005–2025, 7625 days, read once). S1 passed in 3/3 cities and S2 passed: the mean long-horizon gain was +0.0757

TABLE VII. Step 6 (validation only): mean pooled validation R^2 over the eight development datasets. QRC: crotonic-9, $\tau = 30\,\mathrm{ms}$; readouts with their feature count D. Base rows in bold.

part	condition	D	mean val. R^2
A	$\mathbf{0.35}\arcsin\sqrt{\mathbf{s}}$ (final)	653	**0.7060**
	$\arcsin s$ (Ref. [22])	653	0.6903
	$\arcsin\sqrt{s}$ $(g = 1)$	653	0.6798
B	**proton FID** (final)	653	**0.7060**
	^{13}C FID	653	0.7046
	proton $\oplus$ ^{13}C FID	1306	0.7083
	proton $\oplus$ $\langle\sigma_z\rangle$	662	0.7058
	proton $\oplus$ populations	1165	0.7068

(threshold +0.0205) and the mean short-horizon gain +0.0185 (no loss) (Tables VIII and IX). A secondary rule—re-selecting g on each city's own validation period—picked $g = 0.5$ in Chennai and Cairo and was worse there than the fixed rule (Cairo: +0.0026, CI $[-0.0041, +0.0102]$ over $g = 1$): a single frozen gain generalized better than per-city re-tuning. Descriptively, the gain closed most of the gap to the tuned ESN without overtaking it (comparable in Cairo, ESN significantly better in Chennai and Moscow).

B. Confirmation 2: the final design

After steps 2–4 left the design unchanged, Confirmation 2 (commit `7965cde`, before the data were downloaded) tested the final design on Sydney, São Paulo and Toronto with the same protocol and two pre-registered criteria: F1, a replication of the gain effect ($\Delta(g{=}0.35 - g{=}1) > 0$ with CI excluding 0 in $\geq 2/3$ cities); and F2, non-inferiority with margin $\delta = 0.02$ in pooled R^2, separately for each baseline $b \in \{$tuned ESN, tuned LSTM, linear AR$\}$: the CI lower bound of $\Delta(\text{final} - b)$ must exceed -0.02 in $\geq 2/3$ cities. No superiority over the ESN would be claimed unless a CI lay wholly above 0.

Disclosure. The margin $\delta = 0.02$ was set with knowledge of the Confirmation 1 results, in which the final design's CI lower bounds against the tuned ESN were -0.0220, -0.0054 and -0.0160; the Confirmation 2 data were untouched when it was set. Applied retrospectively

to Confirmation 1, the same margin would pass against the ESN in 2/3 cities (Chennai fails at -0.0220), against the LSTM in 2/3 (Moscow fails at -0.0213) and against linear AR in 3/3.

Outcome. F1 passed in 3/3 cities (gains $+0.0238$, $+0.0513$, $+0.0257$; long-horizon gains $+0.0465$ to $+0.1035$). F2 passed for every baseline in 3/3 cities. Against the tuned baselines the final design is within 0.009 R^2 of the best model in every city: the tuned ESN remains significantly better in Sydney (-0.0089) and Toronto (-0.0046) and is tied in São Paulo; the tuned LSTM is significantly better only in Sydney (-0.0048); the QRC is significantly better than linear AR in Sydney and São Paulo.

C. Confirmation 3: the kernel readout (not confirmed)

Confirmation 3 (commit `a7cb3e6`, before the data were downloaded) tested the step-5 kernel readout on Lagos, Mexico City and Stockholm. G1 required it to beat the plain ridge readout (CI excluding 0) in $\geq 2/3$ cities; it was significantly *worse* in all three (-0.086, -0.179, -0.007; Table IX). G2, non-inferiority to the tuned ESN given the same kernel readout, passed only in Stockholm. The losses are largest where the test inputs are most shifted (Lagos and Mexico City, $+0.5$ training s.d. in temperature): the kernel extrapolates badly. The kernel readout is withdrawn and the final design is unchanged. Scored descriptively, the final design was non-inferior to the tuned ESN in Mexico City (-0.0040) and Stockholm (-0.0092) and better than the tuned LSTM in all three cities, but in Lagos it was clearly behind the ESN (-0.0491, CI $[-0.0649, -0.0360]$)—the largest gap in any untouched city, and with Chennai (lower bound -0.0220) one of the two cities outside the margin.

Why Lagos. (Post-hoc, descriptive; nine cities.) The Lagos gap grows with horizon, from $+0.016$ (T, $h = 1$) to -0.159 (T, $h = 45$). Lagos temperature anomalies (deviations from the seasonal cycle) are unusually persistent: their 30-day autocorrelation is 0.32 (0.23 after removing the linear trend), against 0.02–0.12 in the other eight cities, and across the nine cities this statistic is the strongest correlate of the QRC$-$ESN gap ($r = -0.93$; driven by Lagos). Slow, small anomalies are exactly what the QRC's shorter memory (Fig. 3) cannot hold, so the Lagos result locates the remaining gap rather than contradicting the others (Fig. 4b).

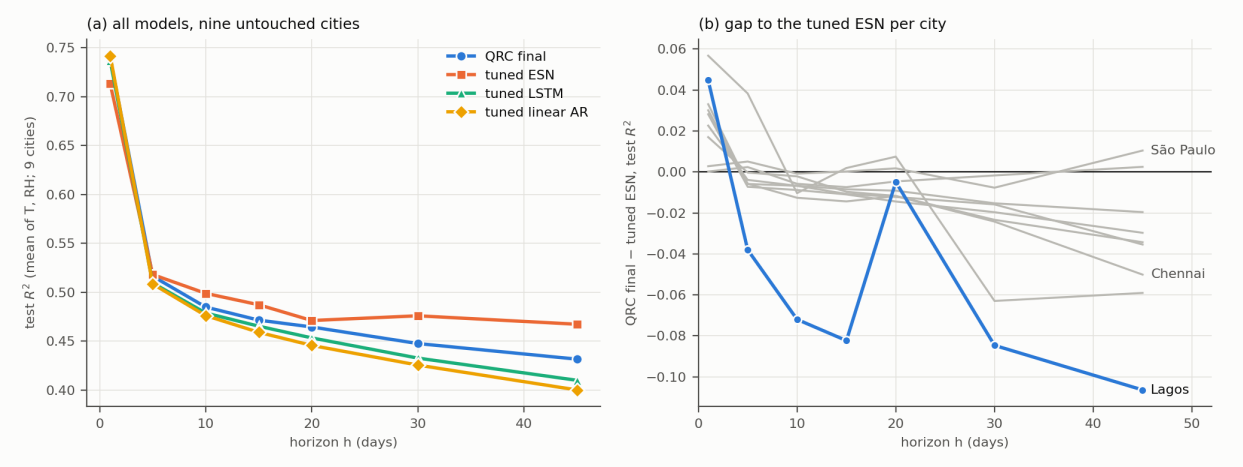


FIG. 4. Test R^2 by forecast horizon on the nine untouched cities (653-bin FID readout, $D = 653$). (a) Mean over cities and the two variables for the final QRC and the tuned baselines: the QRC matches the classical models at short horizons and stays above the LSTM and linear AR, but the tuned ESN pulls ahead beyond about 10 days. (b) QRC minus tuned ESN per city; Lagos (blue) has the largest long-horizon gap.

D. All nine untouched cities

Pooling the descriptive comparisons over all nine confirmation cities (Table X, Fig. 5) gives the summary we stand behind. The final design is close to the tuned ESN but consistently below it (mean $\Delta = -0.0109$; ESN significantly better in 6/9 cities, the QRC in none; within the 0.02 margin in 7/9, failing in Chennai and Lagos). It is ahead of the tuned LSTM (mean +0.0099; 4/9 vs 2/9) and of linear AR (mean +0.0141; 5/9 vs 0/9). The supported claim is *non-inferiority* of a physical QRC to tuned classical models on most untouched real-world forecasting tasks, *not* superiority to the tuned ESN.

VII. PRE-REGISTERED DRIFT TEST

The QRC's failure under input shift (ETTh1, ETTh2) was the clearest weakness of the development benchmark, and step 5 suggested a remedy: re-centre each target on the last observed value and append raw input lags ("level handling"). These are classical features, not reservoir features; the same variant is given to the tuned ESN. The drift test (commit `a4111de`, before the data were downloaded) applied it to two untouched series (Sec. III)

TABLE VIII. Pooled test R^2 on the nine untouched cities (test 2005–2025, 7625 days, read once). QRC: crotonic-9, $\tau = 30\,\text{ms}$, 653-bin FID readout ($D = 653$), regime A; "final" = frozen gain $g = 0.35$, "standard" = $g = 1$. Classical models: mean over three replicates. Confirmation 3 did not run $g = 1$ or the size-matched ESN (—); its kernel-readout scores are in the text.

	city	QRC $g = 1$	QRC final	tuned ESN	ESN ≤ 653	tuned LSTM	linear AR
Conf. 1	Chennai	0.5052	0.5846	0.5973	0.5973	0.5749	0.5567
	Cairo	0.5291	0.5441	0.5462	0.5429	0.5260	0.5179
	Moscow	0.5648	0.5984	0.6073	0.6073	0.6130	0.5971
Conf. 2	Sydney	0.4242	0.4480	0.4569	0.4560	0.4528	0.4383
	São Paulo	0.2309	0.2822	0.2805	0.2746	0.2836	0.2582
	Toronto	0.4537	0.4795	0.4840	0.4831	0.4813	0.4762
Conf. 3	Lagos	—	0.5415	0.5906	—	0.5272	0.5395
	Mexico City	—	0.5102	0.5142	—	0.4579	0.4783
	Stockholm	—	0.5829	0.5921	—	0.5656	0.5819

with one primary criterion, D1: level handling must beat the plain readout (CI excluding 0) on both. Bootstrap blocks are 30 days (exchange) and 168 hours (electricity).

D1 passed on both series (Fig. 6): $+0.058$ $[+0.013, +0.104]$ on the exchange rate, whose test inputs lie 1.26 training s.d. above the training mean, and $+0.010$ $[+0.002, +0.016]$ on electricity, which barely shifts (+0.02 s.d.). The fix works, but it brings no quantum edge. On the exchange rate the QRC with level handling ties persistence (difference -0.00004, CI $[-0.0007, +0.0005]$) and edges past the tuned ESN by $+0.001$: the series is close to a random walk, and level handling lets the readout fall back to the last value. On electricity the QRC stays clearly behind every tuned model (ESN -0.074, LSTM -0.031, linear AR -0.034).

VIII. MECHANISM ON SYNTHETIC TASKS (BRIEF)

Synthetic tasks are not the paper's performance claim; we use them only for ablations that are too costly on multi-decade real series. They use the regime-B learned per-spin encoding (Sec. II I), which needs gradients through the readout, so they use the differentiable physics-

TABLE IX. Paired differences in pooled test R^2, point estimate and 95% moving-block bootstrap CI (30-day blocks, 2000 resamples). First column: the gain effect (S1 / F1); for Confirmation 3, the pre-registered G1, kernel readout minus plain readout. Remaining columns: final design minus each tuned baseline (S3 descriptive in Confirmation 1; F2 non-inferiority, margin −0.02 on the CI lower bound, in Confirmation 2; descriptive in Confirmation 3).

	city	final − standard	final − ESN	final − LSTM	final − AR
Conf. 1	Chennai	+0.0794 [+0.0614, +0.1008]	−0.0127 [−0.0220, −0.0039]	+0.0097 [−0.0020, +0.0225]	+0.0279 [+0.0173, +0.0388]
	Cairo	+0.0150 [+0.0094, +0.0215]	−0.0021 [−0.0054, +0.0010]	+0.0181 [+0.0095, +0.0264]	+0.0262 [+0.0177, +0.0349]
	Moscow	+0.0336 [+0.0250, +0.0432]	−0.0089 [−0.0160, −0.0019]	−0.0146 [−0.0213, −0.0082]	+0.0013 [−0.0041, +0.0067]
Conf. 2	Sydney	+0.0238 [+0.0178, +0.0299]	−0.0089 [−0.0128, −0.0052]	−0.0048 [−0.0083, −0.0014]	+0.0097 [+0.0054, +0.0142]
	São Paulo	+0.0513 [+0.0434, +0.0590]	+0.0017 [−0.0020, +0.0052]	−0.0014 [−0.0061, +0.0034]	+0.0240 [+0.0183, +0.0298]
	Toronto	+0.0257 [+0.0207, +0.0316]	−0.0046 [−0.0080, −0.0015]	−0.0018 [−0.0060, +0.0025]	+0.0033 [−0.0015, +0.0074]
		kernel − plain (G1)			
Conf. 3	Lagos	−0.0862 [−0.1723, −0.0331]	−0.0491 [−0.0649, −0.0360]	+0.0143 [+0.0006, +0.0270]	+0.0021 [−0.0148, +0.0181]
	Mexico City	−0.1788 [−0.2873, −0.1015]	−0.0040 [−0.0197, +0.0123]	+0.0523 [+0.0362, +0.0677]	+0.0319 [+0.0148, +0.0486]
	Stockholm	−0.0066 [−0.0133, −0.0005]	−0.0092 [−0.0144, −0.0044]	+0.0173 [+0.0095, +0.0251]	+0.0010 [−0.0033, +0.0050]

TABLE X. Final design minus baseline over all nine untouched cities (Confirmations 1–3; descriptive). Non-inf.: CI lower bound > -0.02.

baseline	mean Δ	QRC sig. better	baseline sig. better	non-inf.
tuned ESN	−0.0109	0/9	6/9	7/9
tuned LSTM	+0.0099	4/9	2/9	8/9
tuned linear AR	+0.0141	5/9	0/9	9/9

informed line readout of Sec. II G ($D = 2D_{\text{eff}}$, Table I).

On NARMA-2, NARMA-10 [7] and Mackey–Glass [8] ($h = 10$), with a time-contiguous 50/25/25 split, the learned encoding beats the fixed arcsin $\sqrt{s}$ on every seed (Table XI). Re-initializing the reservoir before every step ($\tau \to 0$) removes the gain, so it is carried by cross-step reservoir memory; switching off all couplings ($J = 0$) in the seven-spin dccb7 molecule

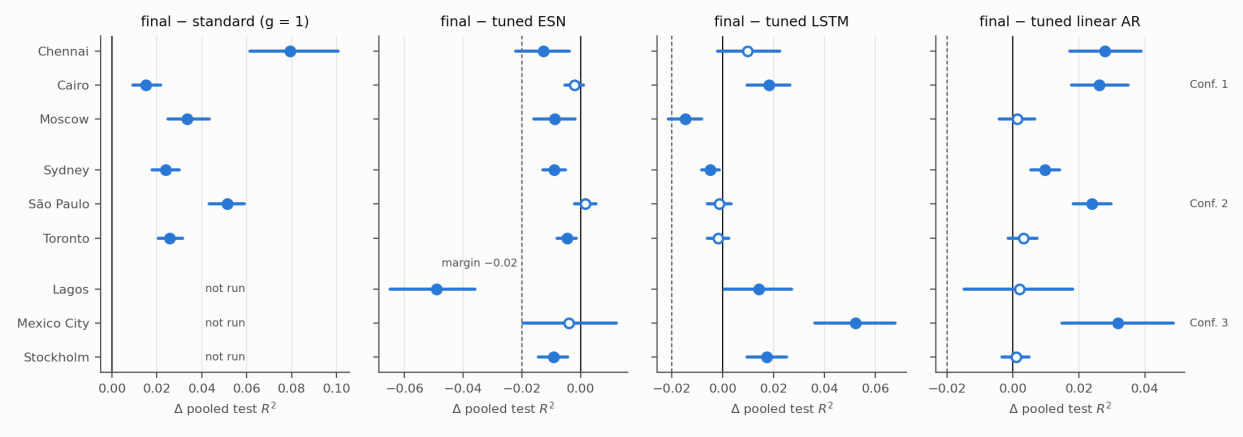


FIG. 5. Paired differences in pooled test R^2 on the nine untouched cities (point and 95% moving-block bootstrap CI; filled: CI excludes 0). Left: the gain effect (final $g = 0.35$ minus standard $g = 1$; not run in Confirmation 3). Others: final design minus each tuned baseline; dashed line, non-inferiority margin -0.02. QRC: 653-bin FID readout, $D = 653$. Values in Table IX.

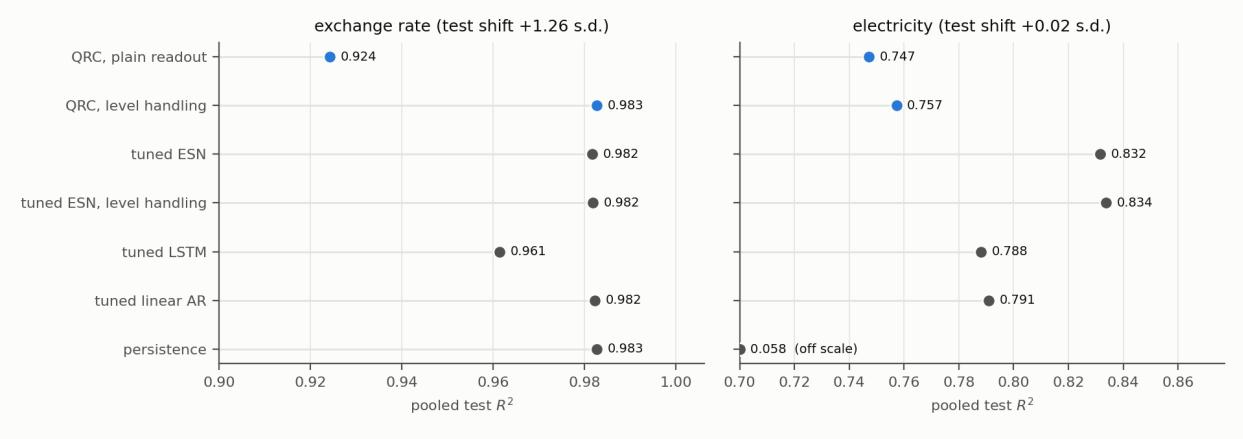


FIG. 6. Drift test, pooled test R^2 on two untouched series (each panel on its own scale). QRC (blue): final design, 653-bin FID readout ($D = 653$), plain or with level handling (last-value re-centring and raw input lags; classical features). Tuned ESN: mean of three replicates. Shift: mean test input in training s.d.

makes it ∼38–126× worse, so the spin–spin coupling carries most of the computation. A tuned LSTM is beaten on NARMA-10 and Mackey–Glass, but a fully tuned ESN (spectral radius, leak, input scale and penalty; also size-matched) is ∼11–830× better on every task. An ESN tuned over size alone had made the QRC look competitive; tuning removed that impression, which is why the real-world study tunes every baseline.

TABLE XI. Synthetic tasks, abstract six-spin Ising reservoir, reduced-FID readout ($D = 322$): geometric-mean test-NMSE ratio of the learned encoding to each comparator (< 1 favours the learned QRC); in parentheses, seeds won by the learned QRC / paired data seeds.

task	vs arcsin	vs $\tau \to 0$	vs tuned LSTM	vs tuned ESN
NARMA-2	0.053 (5/5)	0.0049 (3/3)	1.42 (1/3)	446 (0/3)
NARMA-10	0.41 (5/5)	0.12 (3/3)	0.49 (3/3)	11.5 (0/3)
MG h=10	0.053 (5/5)	—	0.58 (3/3)	832 (0/3)

On real data the learned encoding improved on arcsin at $g = 1$ on 8 of 9 development datasets (Table IV), but the fixed gain $g = 0.35$ captured that improvement and learning on top of it added nothing (Sec. V D). The real-world design therefore keeps the physically simplest lever—how far each input pulse rotates the stored state.

IX. THREATS TO VALIDITY

1. *Simulation only.* All dynamics are simulated (exact Lindblad, measured crotonic parameters). The simulation reads the FID from a copy of the state each step; on hardware the readout disturbs the ensemble, so an experiment must re-run the input history per readout point or use weak readout, at a real cost in experiment time.

2. *One reservoir, one task family in confirmation.* The real-world results use a single molecule (crotonic-9). All nine confirmation cities are ERA5 daily temperature and humidity with one protocol; ERA5 is reanalysis, smoother than station data. Outside daily weather the evidence is weaker: the development benchmark shows the QRC behind (hourly data, Santa Fe) or failing (ETTh1/ETTh2), and on the untouched electricity series it is behind every tuned model (Sec. VII).

3. *Margin chosen after Confirmation 1.* The non-inferiority margin $\delta = 0.02$ was set knowing the Confirmation 1 CIs (Sec. VI); with a margin of 0.01 Sydney would fail against the ESN (lower bound -0.0128), although F2-ESN would still pass in 2/3 cities. The margin is a judgement of practical equivalence, not a property of the data.

4. *Seen development test sets.* The development benchmark's test blocks were scored

before the design work; although every design step used validation data only, the concepts were motivated by those results. This is why every confirmation and the drift test use data downloaded only after pre-registration. The Lagos explanation (Sec. VI C) is post-hoc and rests on nine cities.

5. *Uncertainty model.* The bootstrap CIs resample test time for fixed fitted models; they do not include replicate (seed) variance of the ESN and LSTM, which is reported separately (Table IV; replicate s.d. up to 0.0096 in the confirmation cities). The nine per-city comparisons per baseline are not corrected for multiplicity; we report counts, not a pooled test.

6. *Distribution shift.* Under input drift the QRC's nonlinear features extrapolate badly; bounded transforms did not fix it, and a kernel readout made it worse. Level handling fixes most of it (Sec. VII), but it is a classical addition to the readout, and only one of the two drift-test series is strongly shifted.

7. *Mechanism study.* The ablations use the reduced-FID readout ($D = 200$–344), not the 653-bin standard, mostly on an abstract six-spin Ising model; comparisons with the tuned classical models use three data seeds. They were not run on real-world data.

8. *Six-qubit differentiability wall.* Regime B on the dense propagator is limited to six qubits; the nine-spin and seven-spin campaigns use the block propagator, and larger systems need checkpointing or parameter-shift gradients. The parameter-shift training loop is shown only on a four-spin test reservoir (Fig. 2); at nine spins it costs $2nT$ full reservoir runs per step.

X. CONCLUSION

On real-world forecasting with chronological splits, validation-only development, three pre-registered confirmations on nine untouched cities and a pre-registered drift test, a simulated nine-spin NMR reservoir with the standard 653-bin FID readout is non-inferior to a tuned ESN in 7/9 cities, to a tuned LSTM in 8/9 and to a tuned linear AR model in 9/9, but never better than the tuned ESN, which is significantly ahead in 6/9. The step that made the QRC competitive was physical and simple: reducing the encoding gain to $g = 0.35$

to limit input overwrite. The remaining gap is long memory, largest where slow anomalies dominate (Lagos). Five further pre-registered changes did not help—including reading the long-lived ^{13}C spins, which shows that the memory gap is set by the write, not the readout. The QRC's failure under input shift is fixed by classical level handling, which gives it no edge. We therefore report no quantum advantage. The remaining lever is the write itself: storing inputs on spins that later pulses do not touch (which, with a diagonal Hamiltonian, needs a coherent transfer step), or longer-T_1, more strongly coupled molecules, followed by on-device tests.

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